

# Metamaterial Integrated Photonics for Light Coupling, Beam Steering and Spectral Filtering

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**Abstract** *In this paper we provide an overview of recent developments in metamaterial integrated optics including efficient grating couplers on silicon nitride, optical antennas and phased arrays, as well as polarization-independent Bragg filters. ©2026 The Author(s)*

## Introduction

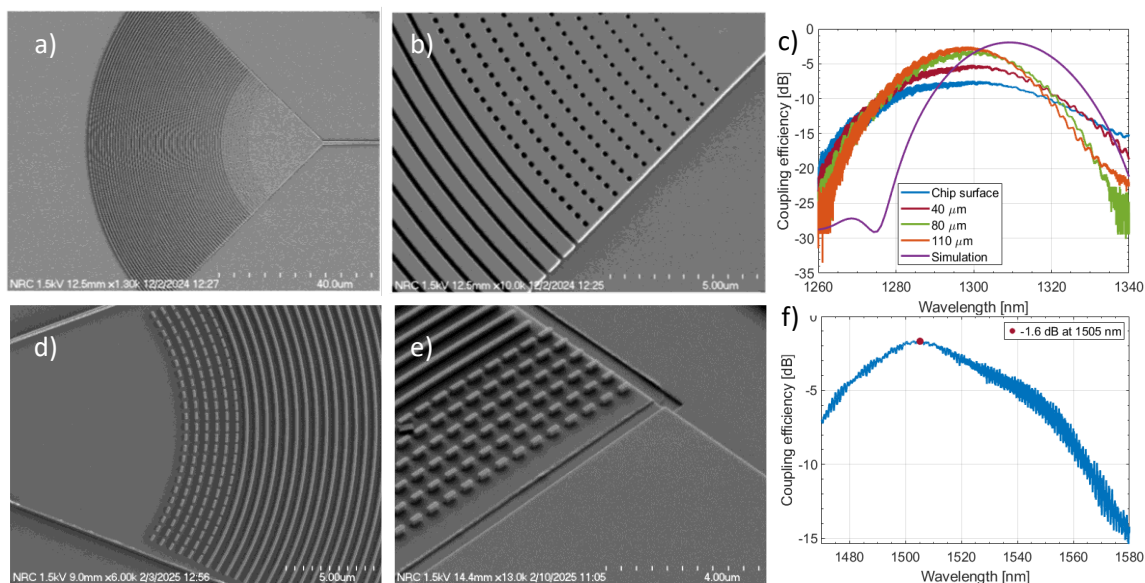
Metamaterial integrated photonics, which brings together the fields of metamaterials and silicon photonics, has been a rapidly growing area of research. By introducing new degrees of freedom in optical device design, metamaterial engineering has resulted in significantly improved performance for many integrated photonic components, such as fiber-chip couplers, polarization control components, waveguide filters, sensors and nonlinear devices [1,2]. Here we highlight some recent developments in application areas that are currently receiving significant attention. This includes the development of low loss components on the silicon nitride (SiN) platform, which is expected to play an important role in the proliferation of integrated quantum photonic devices, and the demonstration of metamaterial engineered optical antennas and phased arrays for beam steering. Finally, we present results of modulated waveguide Bragg filters on the silicon nitride platform designed for a complex and polarization-independent filtering response with single or multiple pass bands.

## Efficient Grating Couplers for Silicon Nitride Waveguides

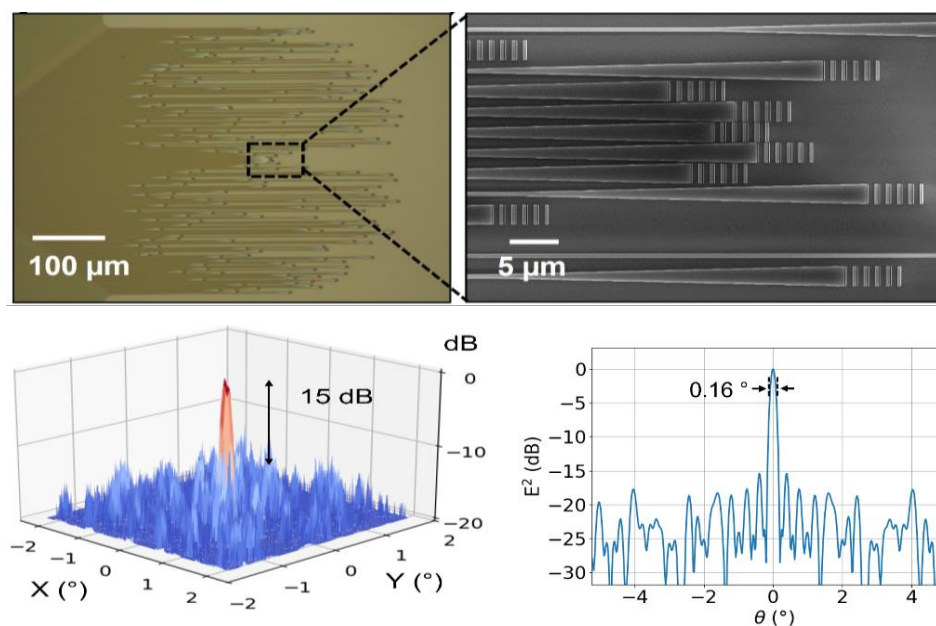
Silicon nitride has emerged as an important waveguide platform for integrated photonics because of its compatibility with CMOS fabrication, its broader spectral transparency range compared to silicon and favourable nonlinear optical properties. Furthermore, silicon nitride

waveguides can have significantly lower propagation loss than silicon waveguides. They are therefore a preferred platform for building quantum photonic devices and circuits. Metamaterial components on SiN have recently been reported. First metamaterial waveguides and ring resonators were designed and fabricated on a 400 nm SiN platform and showed promising performance with Q-factors of  $2 \times 10^5$ , comparable to rings consisting of conventional strip waveguides on the same platform [3].

In Fig. 1a) and b) we show scanning electron micrographs of single etch focusing grating couplers fabricated in a standard 400 nm thick silicon nitride platform. The metamaterial apodized design forms a focused spot matching the optical fibre mode size at a distance of approximately 110  $\mu\text{m}$  above the chip surface. As shown in Fig. 1c, a peak coupling efficiency of -2.5 dB has been achieved. We have recently reported the design of high-efficiency focusing grating couplers for SiN waveguides using a patterned amorphous silicon (a-Si) layer deposited on top of the waveguides. The structure was calculated to have a peak efficiency of -1.3 dB at a wavelength of 1310 nm and a 30 nm bandwidth [4]. First experimental results of a fabricated a-Si grating coupler structure are shown in the coupling curve of Fig. 1f) with a peak coupling efficiency of -1.6 dB at



**Fig. 1:** Metamaterial enhanced grating couplers for silicon nitride: a), b) Focusing apodized grating coupler in 400 nm thick foundry silicon nitride, c) measured fiber-chip coupling curves for several vertical distances between fiber and chip surface. d), e) Grating etched in amorphous silicon overlayer on top of silicon nitride waveguide, e) measured coupling curve.



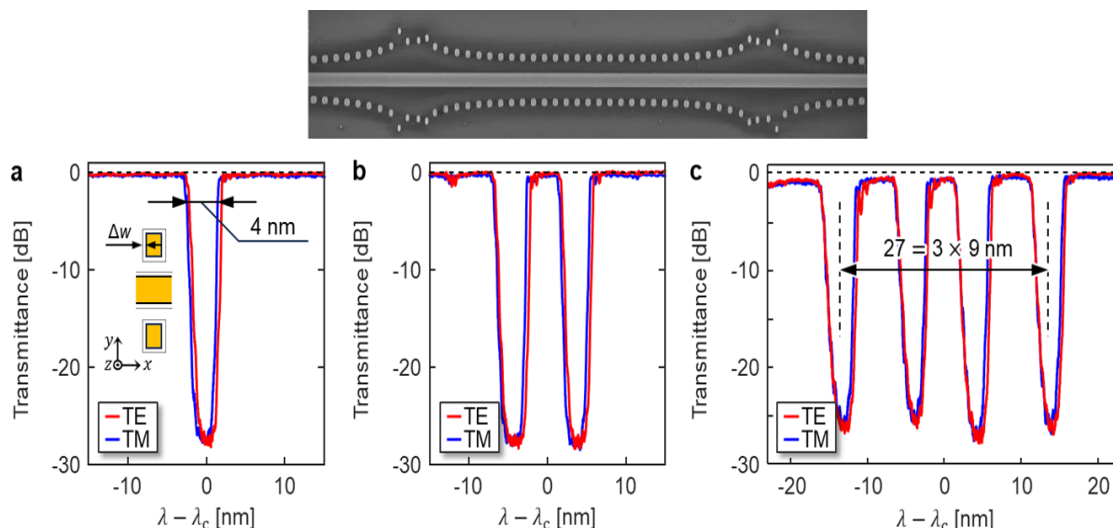
**Fig. 2:** Optical micrograph of optical phased array showing scalable arrangement of grating couplers with fixed vertical spacing and feeder waveguides routed in parallel from left to right (top left), higher magnification view with scanning electron microscopy showing a few individual grating couplers (top right). Bottom left: measured intensity profile of generated beam showing 15 dB background suppression, bottom right: cross-cut view of beam profile.

1505nm wavelength. Further results will be presented at the conference.

### Metamaterial Antennas and Optical Phased Arrays for Beam Steering

Metamaterial engineering has been used for near-field phase front shaping of small footprint silicon optical antennas [5]. The combination of two different pitches alternated at a subwavelength scale in the transverse directions of the antenna is used to control the near-field phase of the vertically emitted light and ultimately results

in a far-field beam profile that overcomes the Fraunhofer limit of the beam width expected from the antenna aperture. Here we present a new deterministic method of spatially arranging the antennas of an optical phase array in way that minimizes the formation of sidelobes and is fully scalable. Figure 2 (top) shows an optical micrograph of such an array as well as a zoomed-in scanning electron micrograph of several individual grating antennas with their feeder waveguides. As can be seen, all feeder waveguides are routed in parallel from left to right each terminating with an



**Fig. 3:** Polarization independent complex Bragg filters. Top: Scanning electron micrograph of fabricated filter structure. Bottom: Measured transmission spectra for transverse electric and transverse magnetic light for three different filter designs: Single band (a), double band (b) and quadruple band (c).

antenna in a design that is scalable to much larger array sizes in a straightforward fashion. The horizontal position of the grating is determined by an algorithm that aims to minimize the formation of lines of antennas in any orientation within the array plane in order to minimize side-lobes. No randomization of antenna positions is used. The bottom panels in Fig. 2 show the measured beam profiles of the 200-antenna array, demonstrating single-beam operation with sidelobe suppression of 15 dB and far-field beam divergence of  $0.16^\circ$ .

### Polarization-Independent Complex Bragg Grating Filters in Silicon Nitride

Integrated optical filters are essential components of on-chip multiplexers, which play a critical role in enhancing data throughput in networks. Bragg filter based on periodic loading segments physically separated from the waveguide core and interacting with the evanescent field of the waveguide mode have previously been used to achieve ultra-narrowband [6] and complex multi-line [7] filters in silicon waveguides. Here the concept is applied to the silicon nitride platform. In evanescently coupled Bragg filters, the grating strength can be modulated along the waveguide by changing the size and the separation of the individual loading segments. By judiciously designing the grating using both size and separation parameters, it is possible to design complex Bragg filters with polarization independent grating strength [8]. A scanning electron micrograph of a fabricated filter is shown in Fig. 3 (top). As a demonstration of the concept, three different types of filters were designed and fabricated: A single band filter with a bandwidth of 4 nm, as well as dual and four-band designs with the same

bandwidth and a channel spacing of 9 nm. Measured transmission spectra for both transverse electric and transverse magnetic light are shown in Fig. 3a)-c) for the three filter types. Polarization-independent operation is achieved with extinction ratios  $>25$  dB and insertion losses below 1 dB. These results demonstrate that the evanescently modulated Bragg filter geometry allows for the versatile design of on-chip waveguide filters with multiple bands and polarization control.

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